Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu

# LAB REPORT

## CSF3023 – Web-based Application Development

| | |
|---|---|
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| **Semester** | 2025/26-2 |
| **Lab Report** | 2 |
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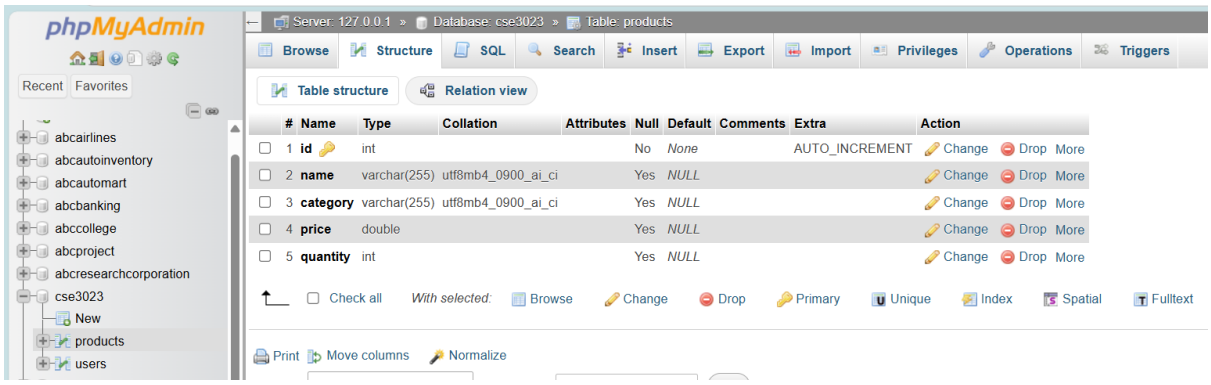
# Lab Exercise

## Database




## JavaBean (Model)

Product.java

```java
package com.lab.model;

/**
 *
 * @author ADMIN
 */
public class Product {
    private int id;
     private String name;
     private String category;
     private double price;
     private int quantity;

     // Constructor kosong (WAJIB)
     public Product() {
     }

     // Constructor dengan ID (untuk update & delete)
     public Product(int id, String name, String category, double price, int quantity) {
         this.id = id;
         this.name = name;
         this.category = category;
         this.price = price;
         this.quantity = quantity;
     }

     // Constructor tanpa ID (untuk insert)
     public Product(String name, String category, double price, int quantity) {
         this.name = name;
         this.category = category;
         this.price = price;
         this.quantity = quantity;
     }
```

```java
    // Getter & Setter
    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

    public double getPrice() {
        return price;
    }

    public void setPrice(double price) {
        this.price = price;
    }

    public int getQuantity() {
        return quantity;
    }

    public void setQuantity(int quantity) {
        this.quantity = quantity;
    }
}
```

# Data Access Object (DAO)

## ProductDAO.java

```java
package com.lab.dao;

import com.lab.model.Product;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

/**
 *
 * @author ADMIN
 */
public class ProductDAO {

    private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "admin";

    // Connect database
    protected Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            connection = DriverManager.getConnection(jdbcURL, jdbcUsername, jdbcPassword);
        } catch (Exception e) {
            e.printStackTrace();
        }
        return connection;
    }

    // CREATE (Insert Product)
    public void insertProduct(Product product) {
        String sql = "INSERT INTO products (name, category, price, quantity) VALUES (?, ?, ?, ?)";
        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {

```

```java
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());

            pstmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }

    // READ (Select by ID)
    public Product selectProduct(int id) {
        Product product = null;
        String sql = "SELECT * FROM products WHERE id = ?";

        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {

            pstmt.setInt(1, id);
            ResultSet rs = pstmt.executeQuery();

            if (rs.next()) {
                String name = rs.getString("name");
                String category = rs.getString("category");
                double price = rs.getDouble("price");
                int quantity = rs.getInt("quantity");

                product = new Product(id, name, category, price, quantity);
            }

        } catch (SQLException e) {
            e.printStackTrace();
        }
        return product;
```

```java
    }

    // READ (Select all)
    public List<Product> selectAllProducts() {
        List<Product> products = new ArrayList<>();
        String sql = "SELECT * FROM products";

        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {

            ResultSet rs = pstmt.executeQuery();

            while (rs.next()) {
                int id = rs.getInt("id");
                String name = rs.getString("name");
                String category = rs.getString("category");
                double price = rs.getDouble("price");
                int quantity = rs.getInt("quantity");

                products.add(new Product(id, name, category, price, quantity));
            }

        } catch (SQLException e) {
            e.printStackTrace();
        }
        return products;
    }

    // UPDATE
    public boolean updateProduct(Product product) {
        boolean rowUpdated = false;
        String sql = "UPDATE products SET name=?, category=?, price=?, quantity=? WHERE id=?";

        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {
```

```java
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.setInt(5, product.getId());

            rowUpdated = pstmt.executeUpdate() > 0;

        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowUpdated;
    }

    // DELETE
    public boolean deleteProduct(int id) {
        boolean rowDeleted = false;
        String sql = "DELETE FROM products WHERE id = ?";

        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {

            pstmt.setInt(1, id);
            rowDeleted = pstmt.executeUpdate() > 0;

        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowDeleted;
    }
}
```

## Servlets (Controller)

InsertProductServlet.java

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author ADMIN
 */
@WebServlet(name = "InsertProductServlet", urlPatterns = {"/InsertProductServlet"})
public class InsertProductServlet extends HttpServlet {

    private ProductDAO productDAO;

    public void init() {
        productDAO = new ProductDAO();
    }

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
```

```java
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet InsertProductServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet InsertProductServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
```

```java
    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product product = new Product(name, category, price, quantity);
        productDAO.insertProduct(product); // DAO only

        response.sendRedirect("ViewProductServlet");
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

# ViewProductServlet.java

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author ADMIN
 */
@WebServlet(name = "ViewProductServlet", urlPatterns = {"/ViewProductServlet"})
public class ViewProductServlet extends HttpServlet {

    private ProductDAO productDAO;

    public void init() {
        productDAO = new ProductDAO();
    }

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
```

```java
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet ViewProductServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet ViewProductServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        List<Product> list = productDAO.selectAllProducts();

```

```java
        out.println("<h2>Product List</h2>");
        out.println("<table border='1'>");
        out.println("<tr>");
        out.println("<th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Quantity</th><th>Actions</th>");
        out.println("</tr>");

        for (Product p : list) {
            out.println("<tr>");
            out.println("<td>" + p.getId() + "</td>");
            out.println("<td>" + p.getName() + "</td>");
            out.println("<td>" + p.getCategory() + "</td>");
            out.println("<td>" + p.getPrice() + "</td>");
            out.println("<td>" + p.getQuantity() + "</td>");

            out.println("<td>");
            out.println("<a href='UpdateProductServlet?id=" + p.getId() + "'>Edit</a> | ");
            out.println("<a href='DeleteProductServlet?id=" + p.getId() + "'>Delete</a>");
            out.println("</td>");

            out.println("</tr>");
        }

        out.println("</table>");
        out.println("<br><a href='add_product.html'>Add New Product</a>");
    }

    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
```

```java
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

# UpdateProductServlet.java

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author ADMIN
 */
@WebServlet(name = "UpdateProductServlet", urlPatterns = {"/UpdateProductServlet"})
public class UpdateProductServlet extends HttpServlet {

    private ProductDAO productDAO;

    public void init() {
        productDAO = new ProductDAO();
    }

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
```

```java
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet UpdateProductServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet UpdateProductServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }

    // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
    /**
     * Handles the HTTP <code>GET</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        int id = Integer.parseInt(request.getParameter("id"));
        Product product = productDAO.selectProduct(id);
```

```java
        if (product != null) {
            out.println("<h2>Update Product</h2>");
            out.println("<form action='UpdateProductServlet' method='POST'>");

            out.println("<input type='hidden' name='id' value='" + product.getId() + "'>");

            out.println("Name: <input type='text' name='name' value='" + product.getName() + "' required><br><br>");
            out.println("Category: <input type='text' name='category' value='" + product.getCategory() + "' required><br><br>");
            out.println("Price: <input type='number' step='0.01' name='price' value='" + product.getPrice() + "' required><br><br>");
            out.println("Quantity: <input type='number' name='quantity' value='" + product.getQuantity() + "' required><br><br>");

            out.println("<input type='submit' value='Update Product'>");
            out.println("</form>");
        }
    }

    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product product = new Product(id, name, category, price, quantity);
```

```java
        Product product = new Product(id, name, category, price, quantity);
        productDAO.updateProduct(product);

        response.sendRedirect("ViewProductServlet");
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

# DeleteProductServlet.java

```java
5   package com.lab.controller;
6
7   import com.lab.dao.ProductDAO;
8   import java.io.IOException;
9   import java.io.IOException;
10  import java.io.PrintWriter;
11  import javax.servlet.ServletException;
12  import javax.servlet.annotation.WebServlet;
13  import javax.servlet.http.HttpServlet;
14  import javax.servlet.http.HttpServletRequest;
15  import javax.servlet.http.HttpServletResponse;
16
17  /**
18   *
19   * @author ADMIN
20   */
21  @WebServlet(name = "DeleteProductServlet", urlPatterns = {"/DeleteProductServlet"})
22  public class DeleteProductServlet extends HttpServlet {
23
24      private ProductDAO productDAO;
25
26      public void init() {
27          productDAO = new ProductDAO();
28      }
29
30      /**
31       * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
32       * methods.
33       *
34       * @param request servlet request
35       * @param response servlet response
36       * @throws ServletException if a servlet-specific error occurs
37       * @throws IOException if an I/O error occurs
38       */
```

```java
39      protected void processRequest(HttpServletRequest request, HttpServletResponse response)
40              throws ServletException, IOException {
41          response.setContentType("text/html;charset=UTF-8");
42          try (PrintWriter out = response.getWriter()) {
43              /* TODO output your page here. You may use following sample code. */
44              out.println("<!DOCTYPE html>");
45              out.println("<html>");
46              out.println("<head>");
47              out.println("<title>Servlet DeleteProductServlet</title>");
48              out.println("</head>");
49              out.println("<body>");
50              out.println("<h1>Servlet DeleteProductServlet at " + request.getContextPath() + "</h1>");
51              out.println("</body>");
52              out.println("</html>");
53          }
54      }
55
56      // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
57      /**
58       * Handles the HTTP <code>GET</code> method.
59       *
60       * @param request servlet request
61       * @param response servlet response
62       * @throws ServletException if a servlet-specific error occurs
63       * @throws IOException if an I/O error occurs
64       */
65      @Override
66      protected void doGet(HttpServletRequest request, HttpServletResponse response)
67              throws ServletException, IOException {
68
69          int id = Integer.parseInt(request.getParameter("id"));
70          productDAO.deleteProduct(id);
71
72          response.sendRedirect("ViewProductServlet");
73      }
```

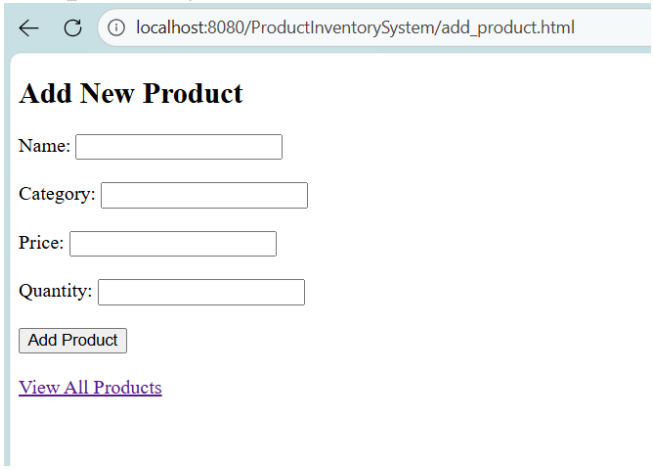
```java
    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

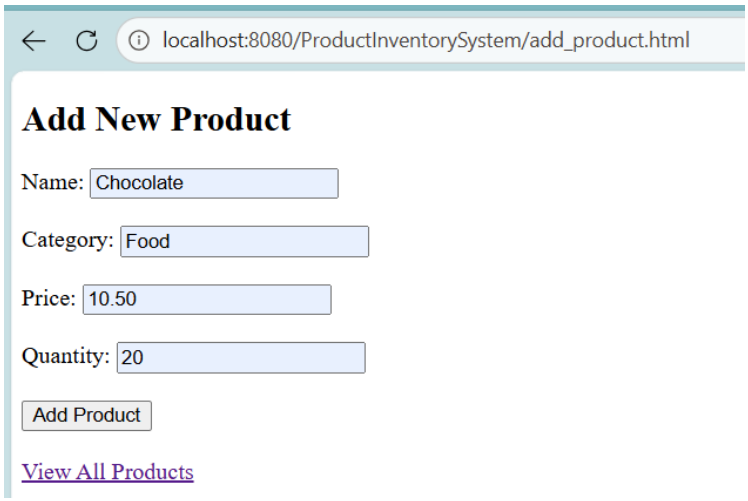
## User Interface (View)

add_product.html

```html
<!DOCTYPE html>
<!--
Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
Click nbfs://nbhost/SystemFileSystem/Templates/JSP Servlet/Html.html to edit this template
-->
<html>
    <head>
        <title>Add Product</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
    </head>
    <body>
        <h2>Add New Product</h2>
        <form action="InsertProductServlet" method="post">
            Name: <input type="text" name="name" required><br><br>
            Category: <input type="text" name="category" required><br><br>
            Price: <input type="number" step="0.01" name="price" required><br><br>
            Quantity: <input type="number" name="quantity" required><br><br>
            <input type="submit" value="Add Product">
        </form>
        <br>
        <a href="ViewProductServlet">View All Products</a>
    </body>
</html>
```

## Output Project



localhost:8080/ProductInventorySystem/ViewProductServlet

# Product List

| ID | Name | Category | Price | Quantity | Actions |
|---|---|---|---|---|---|
| 1 | Chocolate | Food | 10.5 | 20 | Edit \| Delete |

Add New Product

localhost:8080/ProductInventorySystem/ViewProductServlet

# Product List

| ID | Name | Category | Price | Quantity | Actions |
|---|---|---|---|---|---|
| 1 | Chocolate | Food | 10.5 | 20 | Edit \| Delete |
| 2 | Laptop | Electronics | 3500.0 | 6 | Edit \| Delete |

Add New Product

localhost:8080/ProductInventorySystem/UpdateProductServlet?id=1

# Update Product

Name: Chocolate

Category: Food

Price: 10.5

Quantity: 10

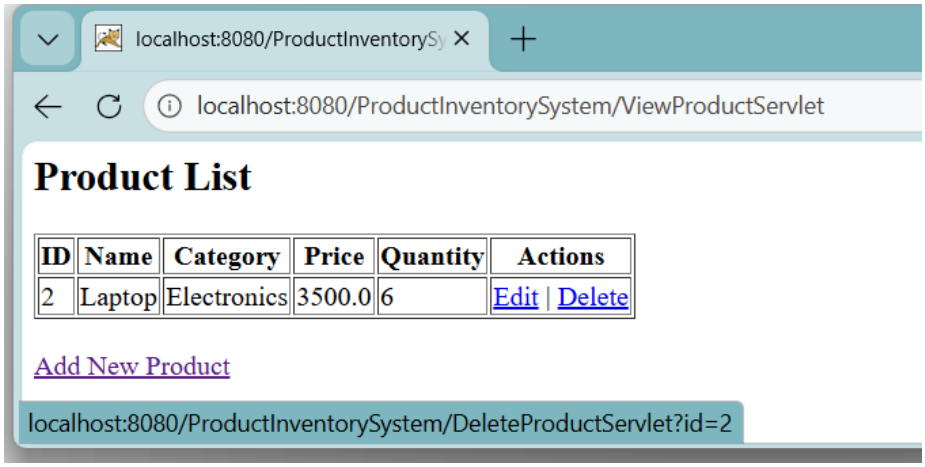
Update Product

localhost:8080/ProductInventorySystem/ViewProductServlet

# Product List

| ID | Name | Category | Price | Quantity | Actions |
|---|---|---|---|---|---|
| 1 | Chocolate | Food | 10.5 | 10 | Edit \| Delete |
| 2 | Laptop | Electronics | 3500.0 | 6 | Edit \| Delete |

Add New Product


localhost:8080/ProductInventorySy
localhost:8080/ProductInventorySystem/ViewProductServlet
Product List
ID Name Category Price Quantity Actions
2 Laptop Electronics 3500.0 6 Edit | Delete
Add New Product
localhost:8080/ProductInventorySystem/DeleteProductServlet?id=2